Multi-Omics Integration Tools and Algorithms

Statistical and correlation-based methods

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Pearson Correlation Coefficient (PCC)

Goal

PCC is used for assessing **the strength and direction of the relationship between two variables.**

Input

Two vectors of numbers with equal sample sizes, for example, a gene expression vector (transcriptomics) and its corresponding protein/metabolite abundance vector, to investigate gene–protein associations and metabolite–enzyme interactions.

Method:

Description

This method shows the relationship and pattern of variables in **normal distribution and linear data**.

Mathematical concept

Its formula is

$$r = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

x and y = the two variables being compared

In this formula, the covariance of variables will be calculated and normalized by the standard deviation. This is unaffected by the unit of measurement.

Output

This method gives us Pearson's coefficient r, which represents the following information:

$0 < r \leq 1$ positive relationship

$-1 \leq r < 0$ negative relationship

Moreover, to test its statistical significance, we can use it with **p-values.**

Interpretation

Useful for analyzing **differential expression of genes, proteins, and metabolites**.

Enables investigation of **correlations between transcriptomic and proteomic data.**

Helps identify **molecular regulatory pathways linking correlated genes and proteins.**

Can reveal time **delays between RNA transcription and resulting protein production**.

Can be combined with **Procrustes statistical shape analysis** to align datasets into a common coordinate space.

Procrustes operations **(scaling, rotation, translation)** remove non-biological structural differences.

Helps detect true geometric similarity and biological correspondence between datasets.

Spearman's Rank Correlation

Goals

Spearman's is used for assessing the **strength of the relationship between two variables** by showing the **ranks** and order of correlations.

Input

We have **Two variables** as input data.

For example: Comparing the value for the **expression of one gene in transcriptomics vs its protein abundance**.

n = number of samples or observations

X=[x_1,x_2,…,x_n]

Y=[y_1,y_2,…,y_n]

Method

These methods measure the strength and direction of the monotonic relationship between two continuous variables, and it is denoted by ρ.

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$$

di = the difference in the ranks of each pair of variables

n = **total number of observations.**

Then, based on values, the rank will be created, which gives rank **1 to the smallest value** and rank 2 to the second smallest one, and so on. After that, the differences between these ranks are squared and summed up.

Output

Provide correlation coefficient ρ values **between -1 and +1** that demonstrate the relationship between variables.

positive value = represents positive relationship

negative value = represents negative relationship

Zero represents = no association

Interoperation

- Integrates multiple omics datasets to **analyze differential expression of genes, proteins, and metabolites.**
- Examines **correlations between transcriptomic and proteomic profiles**.
- **Identifies molecular regulatory pathways** connecting correlated genes and proteins.
- Evaluates pairwise correlations between:
  - Differentially expressed proteins (DEPs) and differential metabolites,
  - Differentially expressed genes (DEGs) and differentially expressed miRNAs,
  - DEPs and DEGs.
- Allows comparison across several datasets to identify **shared biological relationships.**
- Can rank and **display top correlations** (e.g., top 100 strongest correlations).

RV coefficient

Goal

Stands for the **multivariate generalization of the (squared) Pearson correlation coefficient, which is used for entire matrices.** It means this method assess **overall similarity** between **two (multivariate) omics datasets** instead of comparing only two variables (spearman). This method is good for nonlinear data.

Input Data

Two **multivariate matrices**

X = $n \times p$ matrix (n samples, p features → e.g., genes)

Y = $n \times q$ matrix (n samples, q features → e.g., proteins)

Method

$$\mathrm{RV}(\mathrm{X},\mathrm{Y}) = \frac{\mathrm{COVV}(\mathrm{X},\mathrm{Y})}{\sqrt{\mathrm{VAV}(\mathrm{X})\mathrm{VAV}(\mathrm{Y})}}$$

COV: is **scalar covariance** between two multivariate datasets, X and Y. It measures **covariance between two datasets and** compresses **it into a single number**.

$$COVV(X,Y) = Tr(\Sigma_{XY}\Sigma_{YX})$$

ΣXY = covariance X and Y

ΣYX = transpose covariance Y to X matrix

Multiplying the matrices to summarizes this relationship between X and Y

Trace (Tr) is used to extracts **the important diagonal values**, and it summed into **one number**

Finally, we can COVV that gives us one single value that represents the shared structure between X and Y.

VAV is **scalar variance of dataset X**.

It measures the **overall amount of variation in X datasets**, which is summarized in **one single value**.

$$VAV(X) = Tr(\Sigma_{XX}^2)$$

ΣXX: Covariance matrix of X with itself.

For having stronger variances, the matrix is squared.

And then Trace extracts the overall internal variability to a scalar.

Finally, **VAV gives one number representing total variance inside X.**

That can be done for Y datasets too.

$$VAV(Y) = Tr(\Sigma_{YY}^2)$$

Output

It takes values **between 0 and 1. +1** shows very high similarity between datasets, and 0 shows no similarity. It measures the global similarity between two datasets.

Interoperation

So, **Spearman correlation analysis is good for similarity between two variables**, and it is good, for example, for comparing differential gene expression or proteins, while the RV coefficient is good **for analyzing two whole datasets, for example, genomics datasets vs. transcriptomics datasets**, and useful for **multi-omics integration analysis**, for example, investigating the strength of similarities between two omics data sets, genomics vs. transcriptomics.

Canonical Correlation Analysis (CCA) and its regularized form, rCCA

Goal

CCA and rCCA are both **integrative** and **dimensionality reduction methods** that **emphasize correlations between two omics datasets**. rCCA is CCA's variant, and it is particularly for **high-dimensional or multicollinear data**.

Input

CCA two omics datasets and rCCA high-dimensional omics datasets.

X = n×p

Y= n×q

n represents the number of samples

p or q = shows that data for example proteomics or metabolomics

Method

Description

CCA is an unsupervised method that investigates the relationship between two omics datasets through the calculation of canonical variates, a linear combination **that demonstrates maximized correlation between the two novel and simpler components.**

This method works robustly when the ratio of variable to number of samples is less than 1. Nevertheless, we need to use other methods, such as rCCA, **that use ridge (or l2 penalty)** and **shrinkage strategies to handle high-dimensional or collinear data**. This penalty makes diagonals of omics matrices invertible, which effectively addresses the collinearity issue in high-dimensional omics datasets.

Moreover, rCCA can represent a **relevance network based s**olely on pairs of variables that belong to different datasets.

This method is available in the matrix correlation coefficients (MCC) module in Python.

Mathematical concept

We have two datasets, X and Y, and their variables, p and q, respectively. In this method, we want to find their conical variables, **which are U and V, respectively**.

It aims to find a new linear combination **(a and b vectors in this formula)** of variables in numbers X and Y to find the way they have **maximized correlation between U and V.**

U=a_1 X_1+a_2 X_2+⋯+a_p X_p

V=b_1 Y_1+b_2 Y_2+⋯+b_q Y_q

This maximization approach is called the canonical correlation coefficient, which is accessed by maximizing the covariance of the matrix of X, Y, and between X and Y.

$$\rho = \max_{a,b} \operatorname{corr}(U, V) = \max_{a,b} \frac{a^T \Sigma_{XY} b}{\sqrt{a^T \Sigma_{XX} a \; b^T \Sigma_{YY} b}}$$

This maximizing performance has a constraint that states that the variance of the canonical variable is equal to 1.

Var(U)=Var(V)=1Var(U)=Var(V)=1

ρ is equal to the square root of the largest eigenvalue λ1.

ρ=√(λ_1)

Output

As mentioned above, this method provides (1) **canonical variates**, a new component that represents the dataset in a lower dimension and maximizes correlation, and (2) **canonical correlation coefficients** that represent the correlation strength between canonical variate pairs. By using rCCA, we can access a (**3) graphical network representation** of variable connections among different datasets.

Interpretation

To find, for example, which **patterns of genes, proteins, or metabolites change together across samples and the strength of this relationship**.

In studies, rCCA has represented the most significant correlations through relevance networks, and it also **recognized nodes with high connection, which could have a strong impact on diseases.**

Advanced Correlation Measures and Network Extensions

Correlation networks

Goal

This method facilitates visualization and analysis of complex relationships of omics datasets to identify **highly interconnected components** and their **roles through biological systems**, with **assessment of pairwise associations' strength.**

Input

Numeric data from one or multiple omics datasets

Method

Represent pairwise associations in a **graphical correlation network,** where **nodes represent genes, proteins, metabolites, or other biological entities, and edges are built using correlation thresholds (for example, R2 or p-value). The correlation method that is used here includes these types: Spearman, Pearson, and RV Coefficient.**

Output

Network graph, Network modules, Correlation matrix, Hub genes or proteins, Biological subnetworks (demonstrates pathway-level interpretation)

Interoperation

It can represent many valuable biological insights with finding **strong or weak correlations, modules of genes/proteins/metabolites, hub nodes, potential biomarkers or therapeutic targets, and cross-omics edges.**

Weighted Gene Correlation Network Analysis (WGCNA)

Goal

This method helps us to understand the transcriptomic relationship of genes as a system that includes the whole dataset of genes, not just the pairwise correlation, and to find clusters of co-expressed and highly correlated genes as a module.

Input

The WGCNA input usually is a matrix of data that includes the gene expression of each sample. Gene expression matrix (genes × samples).

Method

This method consists of 4 steps:

Step 1: In the first step, **weighted scale-free gene network correlation** networks are built. In this network, **strong correlations are emphasized**, and the weaker or spurious ones are reduced.

Step 2: In the next step, **co-expression modules are identified.** Modules are determined by **hierarchical clustering** on the gene correlation network data.

Step 3: After that, this correlation network information is **simplified by assigning eigengenes to modules**, which is the first component of a principal component analysis. These techniques enable us to analyze the correlation between modules and identify common expression behaviors to identify correlation with phenotypes.

Step 4: Finally**, hub genes** as candidates that have a key role **in certain phenotypes are determined.**

This method is available in an R environment.

Output

- Modules of co-expressed genes
- Module of eigengenes
- Module–trait correlations
- Hub genes
- Biological pathway insights

Interoperation

In one study, WGCNA uncovered **associations between genes/proteins and metabolite modules**. Gene correlation network analysis **provides valuable insights about shared gene regulation and common molecular pathways at the cellular and intercellular levels.** WGCNA can help us find **candidate biomarkers or novel therapeutic targets**. Moreover, WGCNA has been applied successfully in many studies, e.g., cancer, mouse genetics, yeast genetics, and analysis of brain imaging data.

xMWAS

Goals

XMWAS performs **correlation and multivariate analyses** to **integrate two or more omics datasets** to provide **network visualization, clustering, and differential network analysis of data** from biochemical and phenotypic assays.

Input

Two or more omics-type datasets (e.g., transcriptome, metabolome, proteome) on the same set of samples (its matrix form is samples × variables).

Method

Correlation coefficients are calculated by Partial Least Squares (PLS) components and regression coefficients, and this information is used to build a multi-data integrative network graph. The nodes are connected using edges that meet the requirements in terms of association score and statistical significance. After that, **communities that include clusters of highly interconnected nodes,** which are detected using a **multilevel community detection algorithm**, are created. This method **optimizes modularity,** a measurement that represents the higher strength of connection within a community compared to outside of communities. The algorithm is **iteratively updated to reassign nodes to maximize modularity.**

In the second phase, a new network is constructed based on nodes and their communities identified in the previous phase. In the next step, the same algorithm used in the first phase calculates the **final network after many iterations to achieve maximum modularity**.

This method is available as an online tool in the R environment.

Output

- **An integrative network graphs**
- **Clusters (communities) of highly connected features**
- **Differential network analysis results**
- **Lists of important nodes**

Interpretation

The xMWAS method uncovers omics interconnections by identifying the biological pathways that are **detected in highly correlated communities (e.g., identifying a clear pathophysiological pathway).**

What is the Partial Least Squares method (PLS)?

Partial Least Squares (PLS) regression is a statistical method and machine learning that models the relationship between **independent variables (X) and dependent variables (Y) by projecting them into a new space, which is a latent space.** To elaborate on this, it **extracts components that maximize the shared variance (covariance) between X and Y.** This method is useful for multicollinearity data or data with numerous predictors. **This helps in identifying hidden structures and improving prediction in complex datasets.**

Similarity Network Fusion (SNF)

Goal

A graph-based correlation approach **that identifies clusters and relationships for patient omics data according to their omics data profile,** and studies demonstrated this method higher classification performance comparing to other method**. This technique represents a fused network based on the patient's similarity.**

Input

Multi-omics datasets (gene expression matrix, proteomics, miRNA, and RNA expression) of samples (e.g., patients).

Attribute of data sets:

columns represent samples and rows represent features

This information needs to be numerical & appropriately normalized

Method

Description

In this network, nodes **represent patients, and their distance is measured by statistical correlation or other distance-measuring tools such as Euclidean distance.**

After building the **network of similarity of patients from each omics matrix, these networks will be combined using a nonlinear combination method, leading to graph overlapping.**

SNF techniques for merging networks (network fusion) **use the message-passing theory idea**, and it has **iteratively updated networks** to make the **networks more similar together each time**. In this network edge, **a low weight would be removed unless** it was represented in more than one or two networks.

This method is available in MATLAB and R and Python environments.

Spectral clustering SNF: **It identifies molecular subtypes of disease.** After fusing, network spectral clustering is implemented in the network, which **uses eigenvalues instead of patients as nodes** to **project the subject to lower-dimensional space**; this **makes clustering easier.**

Mathematical concept

Step 1: A **K-NN similarity** network will be created based on the omics layer.

If we have n samples and m measurements, we can build network similarity that calculates the **distance between our nodes and weight** (how much our nodes are similar).

In this graph, V is our patients, and E {x1, x2, x3, xn} is our edge that connects patients.

Then the **exponential similarity kernel** is used to determine the weight of the edge.

ρ(x_i,x_j) is the Euclidean distance between patients xi and xj, μ is a **hyperparameter, and εi,j** is used to eliminate the scaling problem.

$$\mathbf{W}(i,j) = \exp(-\frac{\rho^2(x_i,x_j)}{\mu\varepsilon_{i,j}})$$

$$\varepsilon_{i,j} = \frac{\text{mean}(\rho(x_i,N_i)) + \text{mean}(\rho(x_j,N_j)) + \rho(x_i,x_j)}{3}$$

Where mean **(ρ (xi Ni)) is the average value of the distances between xi and each of its neighbors.**

This helps to set similarity values **adjusted based on the local neighborhood of each patient**.

Step 2: Then the similarity matrix will **be normalized to W to get P.**

$$P = D^{-1}W$$

$$\mathbf{P}(i,j) = \begin{cases} \frac{\mathbf{W}(i,j)}{2\Sigma_{k\neq i}\mathbf{W}(i,k)}, j \neq i \\ 1/2, j = i \end{cases}$$

After that, the KNN graph is built; **in this graph, each sample is connected only to its own nearest neighbors based on their similarity.**

$$(i,j) = \begin{cases} \dfrac{\mathbf{W}(i,j)}{\{\Sigma_{k \in N_i} \mathbf{W}(i,k)}, & j \in N_i \\ 0 & \text{otherwise} \end{cases}$$

Step 3: In this step, networks **are fused together iteratively** by using different data types to exchange similarity information.

$$\mathbf{P}_{t+1}^{(1)} = \mathbf{S}^{(1)} \times \mathbf{P}_t^{(2)} \times (\mathbf{S}^{(1)})^T$$
$$\mathbf{P}_{t+1}^{(2)} = \mathbf{S}^{(2)} \times \mathbf{P}_t^{(1)} \times (\mathbf{S}^{(2)})^T$$

And it will be **again normalized.**

In step 4, then the final fused network will be built by calculating the **average of the fused updated networks using this formula**:

$$\mathbf{P}^{(c)} = \frac{\mathbf{P}_t^{(1)} + \mathbf{P}_t^{(2)}}{2}$$

Then this integrated similarity matrix **will be used for spectral clustering or prediction models**.

Output

Provide a (1) **fused correlation network of patient similarities** in omics data sets, in which each patient in scSNF represents a low-dimensioning representation of this network.

SNF is able to (2) **create label and predict new samples comprising** it with the network that is already built of datasets.

Interpretation

The fused network matrix (W_fused) provides these values:

$$W_{ij}^{\text{fused}} \in [0,1]$$

The values that **are close to 1 represent that the samples i and j are very similar across all omics, while those close to 0 show they are different across omics**. And the **high similarity edges demonstrate the consistency among multi-omics datasets.**

This method comprises patients using their omics data on a fused network that represents the correlation between cross-data sets and **performing clustering** to show valuable information such **as molecular subtypes or patient groups and facilitating disease subtype classification using scSNF.**